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Using NJAC Baseball Data to Evaluate Player Awards

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Abstract: NJAC baseball awards the best player and pitcher each year. It is our belief that there is bias in the selection of the players for the award. There are players who deserve recognition in the league but have not gotten it. By using WAR, wins above replacement, we are able to properly evaluate player performance. WAR is a widely agreed-upon way to evaluate a player using a single metric. We modeled WAR, along with other features, to view feature importance. This shows what the voters valued most in awarding players. The models revealed accurate awarding for player of the year, but other features were more impactful in evaluating pitcher performance. Instead of a bias in the voters, there is an imperfect understanding of how to evaluate a pitcher. Division 3 baseball is statistically a few years behind the MLB, as reflected in these metrics. The hope is to modernize the statistics field of NJAC baseball to honor the players who are most deserving.

Keywords: WAR; baseball; NJAC

1. Introduction

The NJAC is widely regarded as one of, if not the most competitive, division three baseball conferences in the country. Year in and year out, multiple teams from the conference are ranked nationally, and players will go on to play professional baseball in some capacity. Every year, the coaches come together to vote on the player and pitcher of the year. Given that these coaches each represent a team in the league, we believe there is bias in the voting for these awards. This could be for many reasons, personal issues with a given coach or player, or a soft spot for either as well. It is very well possible that the bias is not implicit, but rather an imperfect understanding of how to evaluate a player. Using wins above replacement (WAR) as our evaluation metric of performance, we can see which players deserved the award each year and if the coaches got it right. Other features in the model can be seen as the discrepancies in correctly selecting the player for the award. In other words, the "bias" a coach has will be reflected if other features are represented as more important than WAR. The use of logistic regression, support vector machine (SVM), and random forest models is used to view feature importance to understand how the coaches are making their decisions, right or wrong. Players have dedicated their entire lives to baseball. They should be able to earn the awards they deserve from the time and effort spent on the game that they love.

2. Materials and Methods

The data used for the WAR calculations and modeling came from the NJAC baseball statistics archive as well as from each team's respective baseball statistics archive. All data is publicly accessible. For the years 2010 to 2019, the data came from the NJAC website, while the years 2021 to 2025 came from each team's website. It is a coincidence that the storage format changed during 2020, the only year left out within the data due to the lack of games played. This data was collected in Excel files, each year having its own sheet in the file. There were three total files: hitters, pitchers, and fielders, all labeled accordingly. The data was imported and cleaned in the Rmarkdown file,

where all methods were performed. The final Excel file included the final regular-season standings of each NJAC team, along with who won the player of the year and pitcher of the year awards. This file will allow us to integrate the target variable, winning the player or pitcher of the year award.

The data gathered from the sources above included only the statistics. The age of each player was not yet available. The use of a web scrape was necessary to gather the age as well as the position of the players. We did not end up using position, but it is a good variable to have for future work. The stats were joined with the web scrape by the last name of the players, then filtered by first names matching within a few characters. There was minimal hard-coding to account for players of the same name in the same year. It is important to note that not every player in the stats had roster information due to leaving the team mid-season or, for example, Stockton not showing their 2010 roster. The players' data were still used for the WAR calculations, but not in the modeling. These steps were the most time-consuming and tedious. It was vital to accurately pull the information from the websites to ensure validity in the data.

Upon completion of the data gathering and data cleaning, we moved on to calculating our main evaluation statistics, WAR. WAR stands for "Wins Above Replacement" and shows how much better or worse a player is than the league average player. WAR is calculated differently for hitters and pitchers since, by nature, their positions are much different. We used Fangraphs formulas for WAR. For the hitters' WAR[1], we did not include fielding runs and positional adjustment in the formula. The NJAC statistics do not provide ample stats to properly evaluate the defensive value of players. WAR is still scaled correctly, considering the two values are in the numerator of the equation. Beyond those two parts, the formula includes batting runs, base running runs, league adjustment, and replacement runs. Batting and base running runs are straightforward. They offer insight into the value of the player's hitting ability and base stealing. League adjustment is a scaling variable to show the importance of each plate appearance in a given year. Replacement level runs represent how many runs the replacement level player produces compared to the players in the league.

Pitcher WAR, ironically, is much more straightforward. For the pitcher WAR[2], we used the exact formula provided. This formula is built around fielding independent pitching, or FIP. FIP only evaluates a pitcher on outcomes they have control over: strikeouts, walks, hit-by-pitches, and home runs. It takes away balls in play that are too volatile to objectively evaluate. The hands-on part we had in the WAR calculation was obtaining the correct WAR per year. In the MLB, there is 1,000 WAR per year. This is calculated by taking the 162-game season and taking a win percentage of 0.294. This is the winning percentage of a replacement-level team. From that result, you subtract 81 games, half of 162. This final result is multiplied by 30 for each team in the league, and it rounds to 1000. Following the same steps but using the 18-game season of 10 teams, we got a total WAR value of 37. Similar to MLB, we took 57% of the WAR for hitters and 43% of the WAR for pitchers, resulting in 21 and 16 WAR, respectively.

The final addition to the data was two columns representing whether the player had won the award that year or had won the award in a previous year. These were added accordingly, pitcher of the year in the pitchers dataframe and player of the year in the hitters dataframe. The column noting if the player won the award or not will be the target value. The column noting whether the player has won the award before will be a feature in the model. For the steps ahead, we will be doing them on two dataframes: the hitters and the pitchers dataframe.

We moved on to the Exploratory Data Analysis (EDA) after WAR calculations. For the EDA, we narrowed down the features for the modeling. For the pitcher's models, we included winning the award as the target. The features included WAR, innings pitched, games started, appearances, earned run average (ERA), team final standings, year in school, and whether the pitcher has won the award before. Figures 1 and 2 both show some EDA of the pitchers' data. Pitcher WAR in Figure 1 shows a right skew. This is not a concern because of the very limited data that makes it skewed. Inherently, if a pitcher

is performing poorly, they will not pitch. The inverse is true for a pitcher throwing well. By this logic, a right skew seems reasonable. The year in school plot in Figure 2 shows data we believed could be impactful on decision-making. Normal distribution between each age group, with graduates being the smallest naturally. Aside from these two graphs, the distribution of the rest of the data was as expected. In the EDA, we looked at each variable, ensuring normal ranges and distributions. For the purpose of this research, the data does not need to be in any specific distribution aside from WAR. The data reflects baseball, so as long as the distributions and ranges seem reasonable, they are fine to model.

The hitters' EDA took the same steps as the pitchers, with different features by nature. Figure 3 shows the distribution of the hitters' WAR. This is nearly a perfect bell curve, which is expected and desired. It is much easier for a hitter to produce zero WAR than for a pitcher to, therefore evening the distribution slightly. Figure 4 shows the distribution of years in school of the hitters. Slight right skew, as hitters get older, there tend to be fewer of them in the league.

Aside from figures 1 through 4, the distribution of the rest of the data was as expected. In the EDA, we looked at each variable, ensuring normal ranges and distributions. For the purpose of this research, the data is fine as is. The data represents baseball and the human ability to perform. There is no need for any adjustments or removal of any data as long as it has been properly sourced. Most of the time spent on the work came with every step before EDA. Importing data from ten different sites opened small formatting discrepancies that needed to be addressed. By the time EDA was complete, the hardest parts of the project were through.

3. Results

This section contains the results of running the three models and the information they provided. Subsection 3.1 includes the comparison of WAR for each year. 3.2 goes into detail about the models and their results. 3.3 includes all graphs referenced in the paper.

3.1. Comparison of WAR per year

Figures 5 and 6 show a comparison of WAR between the award winner and the player with the largest WAR each year. Figure 5 is the pitchers. For each year, there is a clear discrepancy between the two WAR values. Over the fifteen-year sample size, the voters correctly awarded the best player five times. The other ten instances they missed the mark by a large margin. Taking a look at Figure 6, there is a different visualization. Over the fifteen years, the voters correctly awarded the best hitter five times, but the margin of error is much smaller. Based on the graphs, it seems the voters are better at evaluating hitters more accurately than pitchers.

3.2. Model Results

As shown in Figures 7 and 8, as well as mentioned earlier, we used three models in this research. With the models, we opted to use cross-validation to create better use of the data, given the limited size. An important note is that for modeling, we trimmed the player count to the top 20 players and pitchers each year by WAR. This creates a subjective group of players who are “qualified” to receive the award. Instead of having a formal model evaluation metric, we looked at the feature importance. Feature importance gives insight into which factors were most important in selecting the award winners. For each model, we will evaluate the results for both pitchers and hitters. Overall, each model's results are consistent with one another with mild differences, which show the value of using an ensemble of models.

3.2.1. Logistic Regression

Logistic Regression is noted on both Figure 7 and Figure 8 by a light blue color. The results from this model for pitchers were expected, especially when looking at Figure 5. It was obvious that WAR would not be the most important feature. Earned run average (ERA) and innings pitched both hold great weight in the eyes of the voter. In Figure 8, WAR holds the greatest feature importance, followed by games played. Results are consistent with Figure 6.

3.2.2. Random Forest

Random Forest is noted in both Figure 7 and Figure 8 by a dark blue color. For pitchers in Figure 7, ERA, WAR, and games started proved to be the most important features. WAR came in at third, consistent with Logistic Regression. Figure 8 shows WAR followed by batting average for the hitters. The results reinforce the results from the WAR comparison graphs.

3.2.3. Support Vector Machine (SVM)

SVM is noted in both Figure 7 and Figure 8 with a yellow color. Figure 7 shows similar results to the prior models. ERA and innings pitched lead the way, with WAR being inferior in importance to both. The hitters in Figure 8 get the most diverse importance from SVM. Games played become more important than WAR. The voters hold great importance for the number of games played in a single season. Games played being more important than WAR is an interesting insight we get from SVM.

3.3. Figures

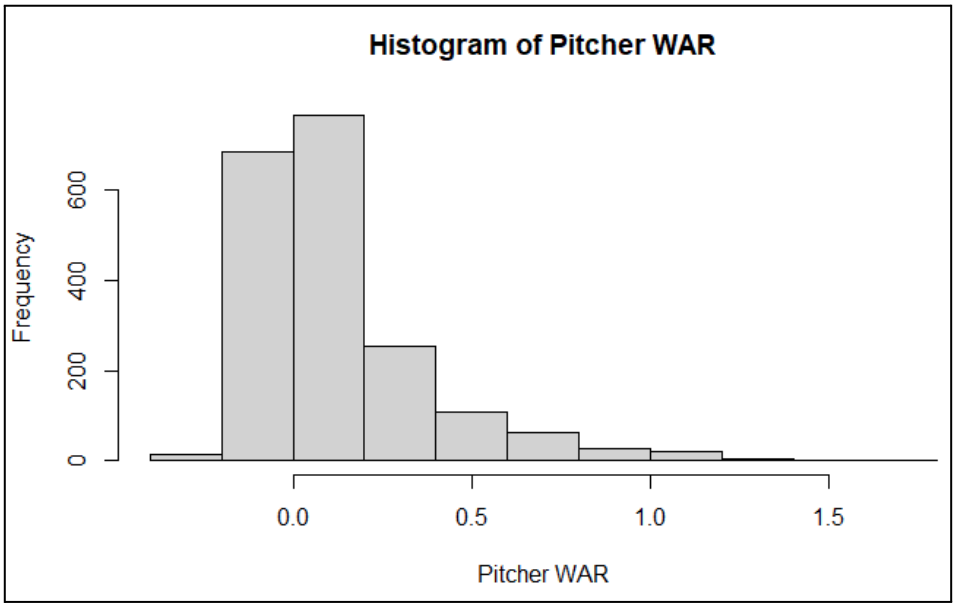


Figure 1. Histogram of pitcher WAR.

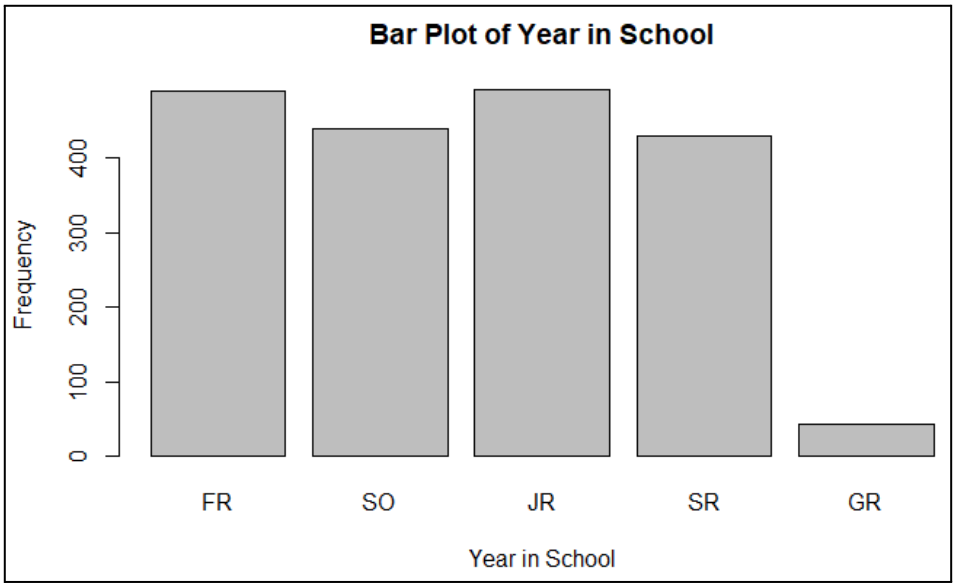


Figure 2. Bar plot of the year in school for pitchers.

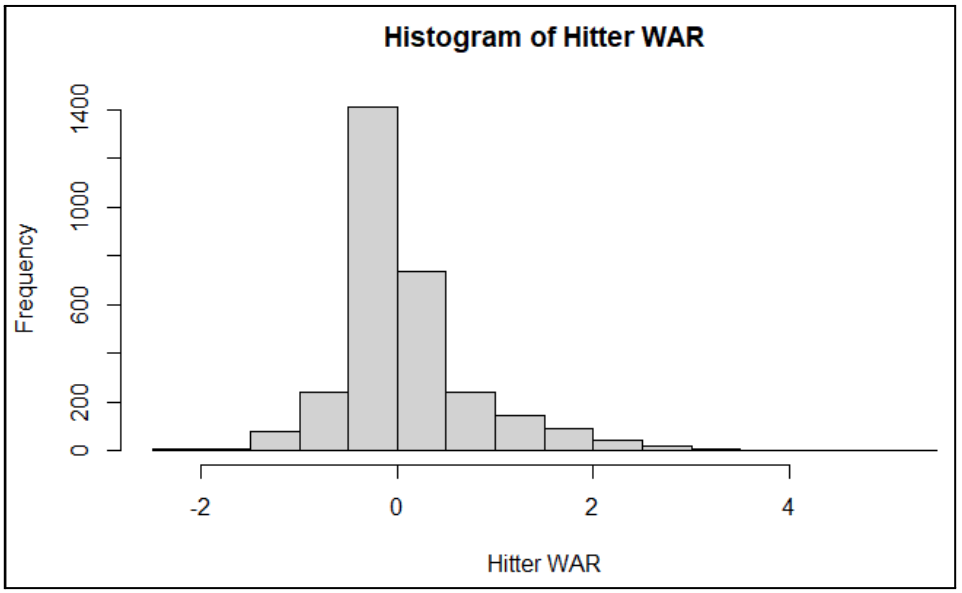


Figure 3. Histogram of hitter WAR.

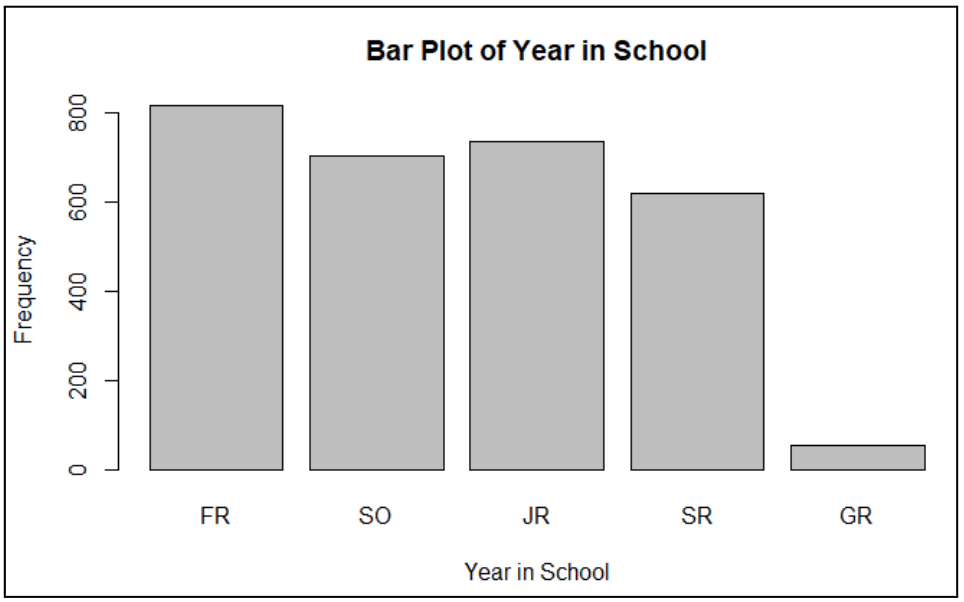


Figure 4. Bar plot of the year in school for hitters.

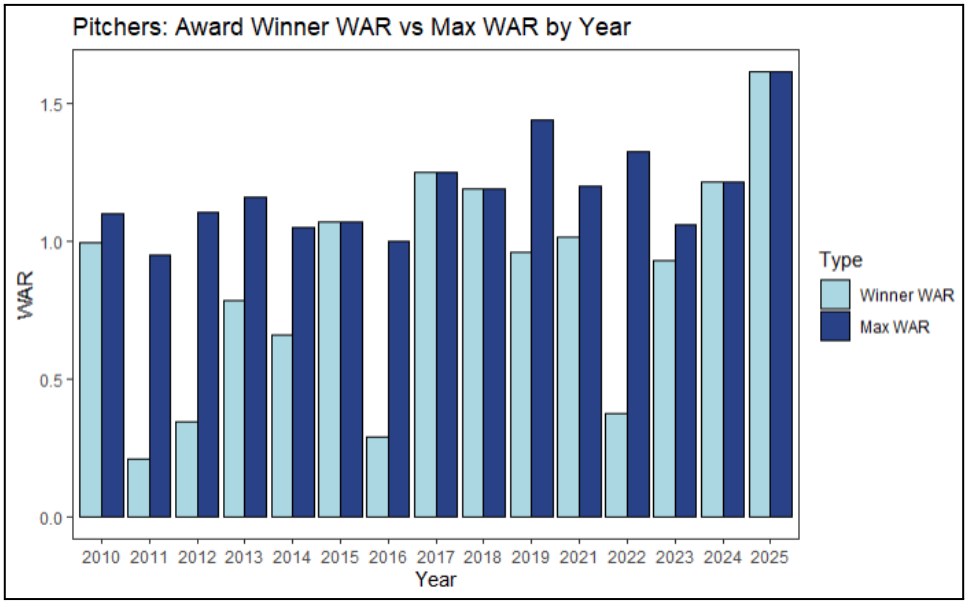


Figure 5. Grouped bar plot for WAR of both award winners and the greatest WAR pitcher per year of the pitchers.

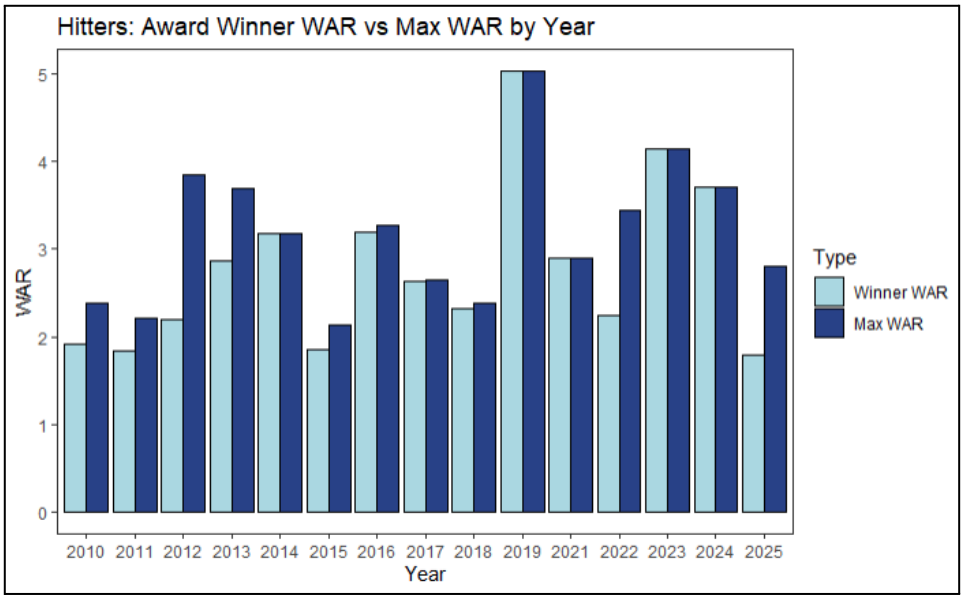


Figure 6. Grouped bar plot for WAR of both award winners and the greatest WAR player per year of the hitters.

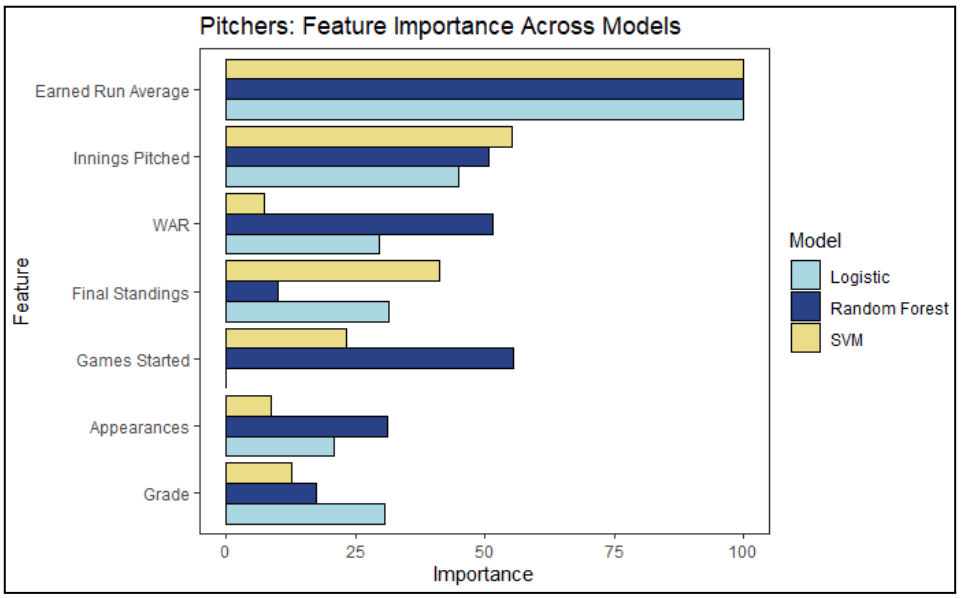


Figure 7. Grouped bar plot for feature importance across all models used (Logistic Regression, Random Forest, and Support Vector Machine [SVM]), for the pitchers.

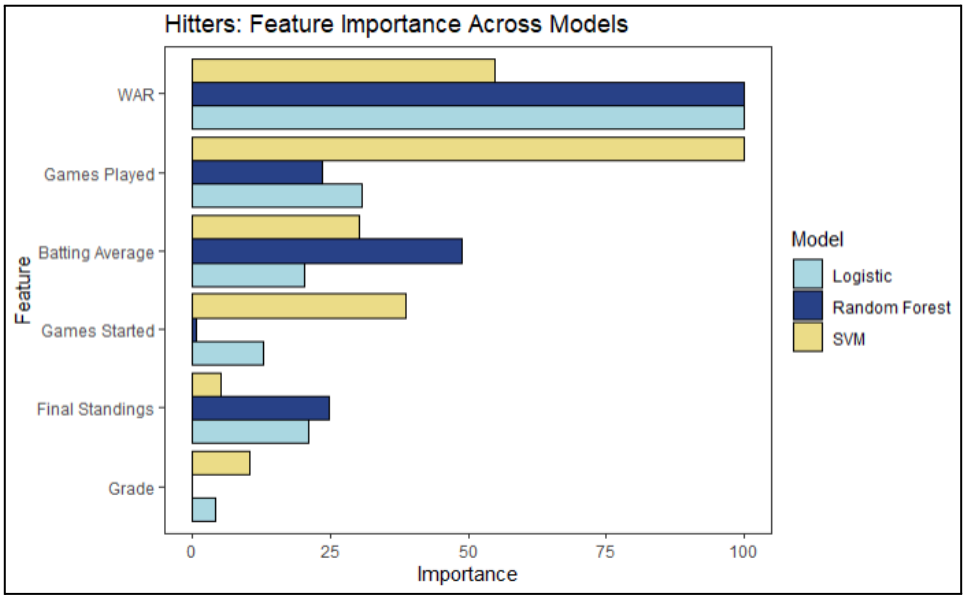


Figure 8. Grouped bar plot for feature importance across all models used (Logistic Regression, Random Forest, and Support Vector Machine [SVM]), for the hitters.

4. Discussion

The hypothesis that originally inspired the project was that there is bias among the coaches voting in these awards. As a player in the league, this comes from a place of frustration since teammates of mine are not receiving awards I believe they deserve. Admittedly, I may be the one holding the most bias going into the research. Be assured that my bias did not affect any results or methods used in this work. The original hypothesis was technically false. My belief has changed to the coaches having an imperfect understanding of how to evaluate players.

The results displayed my new belief. The voters have a difficult time evaluating pitchers. Pitching is a unique position in sports. For years, the main evaluation statistic has been ERA, which is highly subjective. It is at the mercy of the scorekeeper, who can affect it by ruling balls in play as hits or errors. The pitcher has no control over balls in play, yet we, as viewers of the sport, have determined it to be the best way to judge a player's performance. Not shockingly, it is the most important factor in pitchers winning the award. WAR not being as influential on the pitcher awards was expected prior to the creation of Figure 5, but was confirmed by its existence.

The hitters' results were rather shocking. Prior to viewing Figure 6, I expected similar results to the pitchers. I did not expect the voters to be as accurate as they were. I expected features such as batting average, games played, and year in school to hold more value, specifically for the hitters. My understanding after further research and discussion with some voters is the large emphasis placed on power numbers. The ability for the player to get extra base hits and tally runs batted in (RBIs). Over the years, the hitting side of baseball has been filled with statistics to better address the performance of players. With these metrics, viewers of the sport have been better equipped to form their own opinions and views on the players.

My prediction was year in school would be of much larger importance than it did. It simply meant nothing in the models. There have been years where a senior wins the award over a subjectively more deserving underclassman. Of all the results found, this felt the most shocking as I had high hopes that this feature would be impactful.

5. Conclusions

The findings from this research are both reassuring and frustrating. Reassuring that the voters are able to correctly evaluate hitters. Frustrating that this understanding is not applicable to pitchers. I am glad that the feature of team final standing did not play a

role in the awards, as I thought it might. It was a concern of mine that the voters would take the best player from the team that finished first and award that player. While this may be correct some years, it is not correct every year. The implication of this research is awareness of the strengths and weaknesses of the voters' evaluation methods. It is my hope that by identifying the faults in their ways, we may be able to award the most deserving players going forward.

In further research, I would be encouraged to include positional adjustments in the modeling or WAR calculations. While not extremely important, it could give weight to the overall player rather than just the offensive metrics. There would need to be more specific fielding statistics to be able to include the values in WAR, but we could use positions as a grouping metric in the modeling. Another hope with this research is to view the impact of transfer athletes on teams. In the age of the transfer portal, should coaches be putting their focus more on the portal than high school players? For the love of the game, I hope not. This data is widely unexplored, and I believe there is a multitude of information applicable to NJAC baseball.

Appendix A

The discrepancies between my player WAR and the WAR formula from Fangraphs are primarily due to withholding fielding runs and positional adjustments. Fielding runs require Ultimate Zone Rating (UZR), which is a Sabermetric stat. This stat takes a player's ability to correctly position themselves for a batted ball, along with the time it takes to get to the ball. Rather than static statistics, we have for fielding, UZR is tracked with cameras, creating a map of the field. This is way above the head of Division 1 baseball, let alone Division 3. In the same vein, positional adjustments are not practical. NJAC data is not specific enough to calculate the value in runs per position. The final difference lies in the base running runs. We did not use UBR in base running runs (Ultimate Base Running) for non-stolen base running. It represents players ability to go first to third on a base hit or score from second base on a ball to the outfield. The statistics given are not specific enough. While I did try to treat the data as scaled-down MLB data, there are some factors simply not possible due to a lack of data.

References

1. Piper Slowinski. WAR for Position Players | Sabermetrics Library. *Fangraphs* **2012**. Available Online: https://library.fangraphs.com/war/war-position-players/
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